INCH-POUND

MIL-DTL-28748/16
w/AMENDMENT 1
27 February 2019
SUPERSEDING
MIL-DTL-28748/16
12 April 2004

DETAIL SPECIFICATION SHEET

CONNECTORS, ELECTRICAL, HEXAGONAL, RACK AND PANEL, MINIATURE, PLUG-RECEPTACLE, FEMALE SOLDER CUP PIN CONTACTS, SIZE 20

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-DTL-28748.

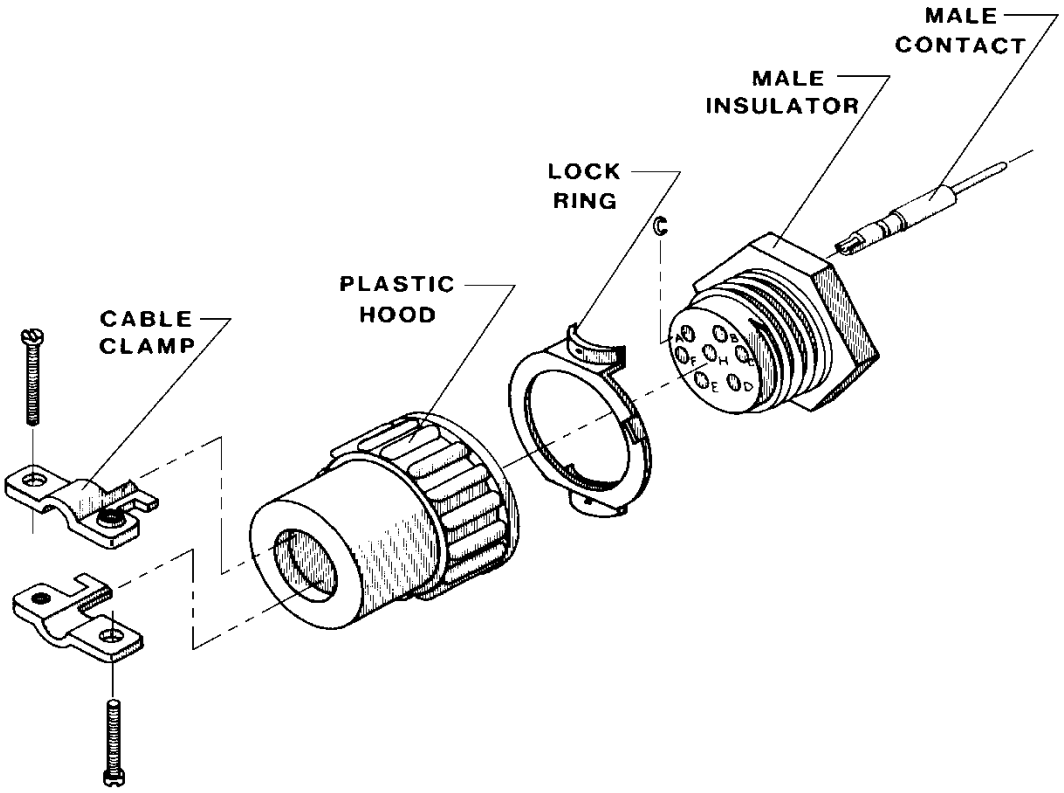
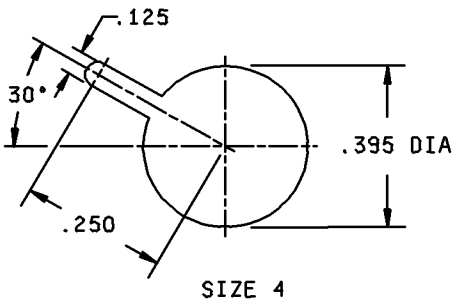


FIGURE 1. Connector.

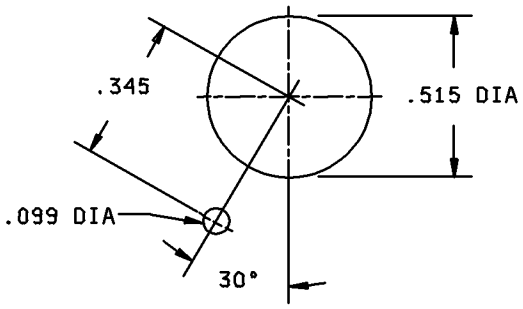
AMSC N/A

FSC 5935

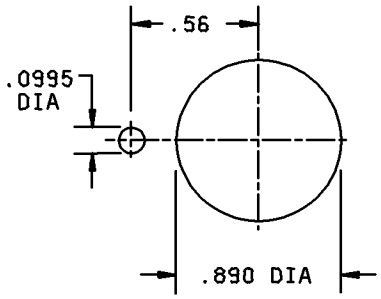
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SIZE 4



SIZE 5,7,9,10



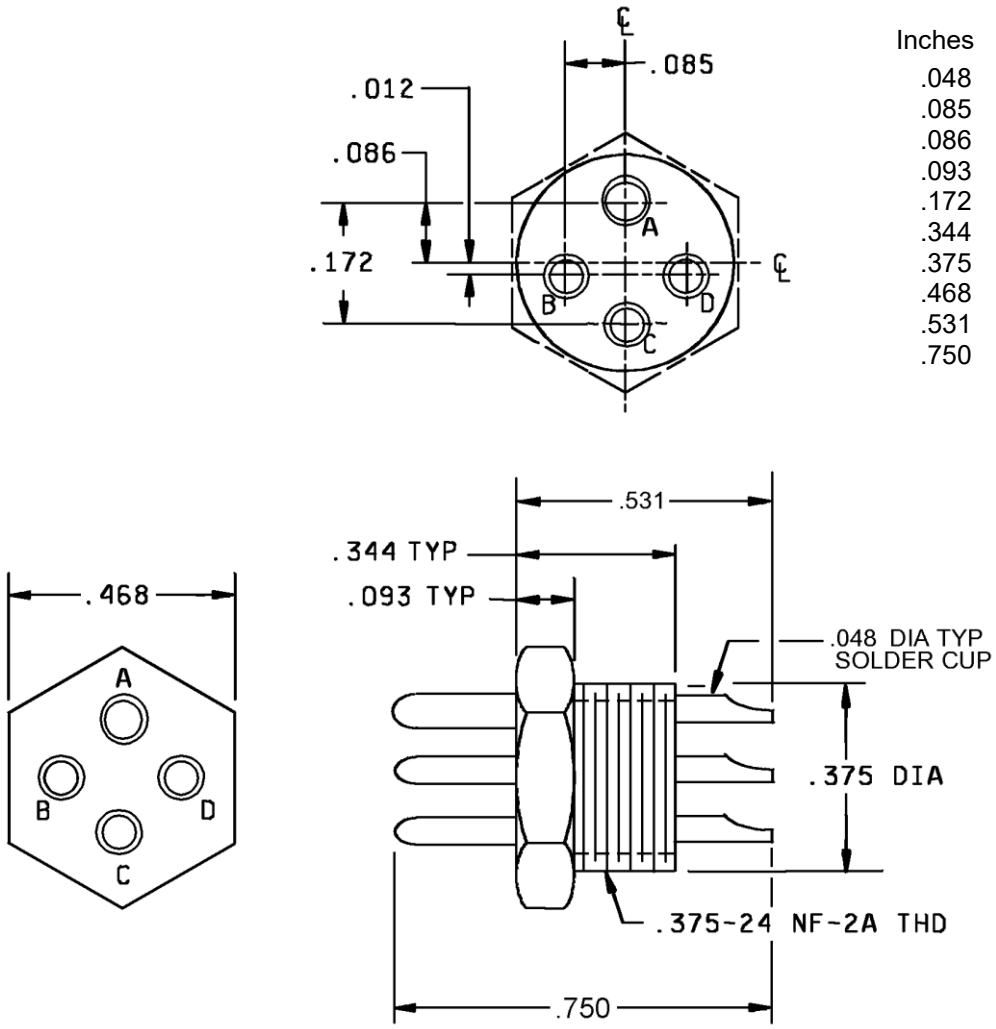
SIZE 12

Inches	mm
.099	2.51
.0995	2.257
.125	3.18
.250	6.35
.345	8.76
.395	10.03
.515	13.08
.56	14.22
.890	22.60

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Panel drilling typical, slot or hole may be rotated to place terminals in desired position.

FIGURE 2. Panel cut-out dimensions (viewed from front of panel).

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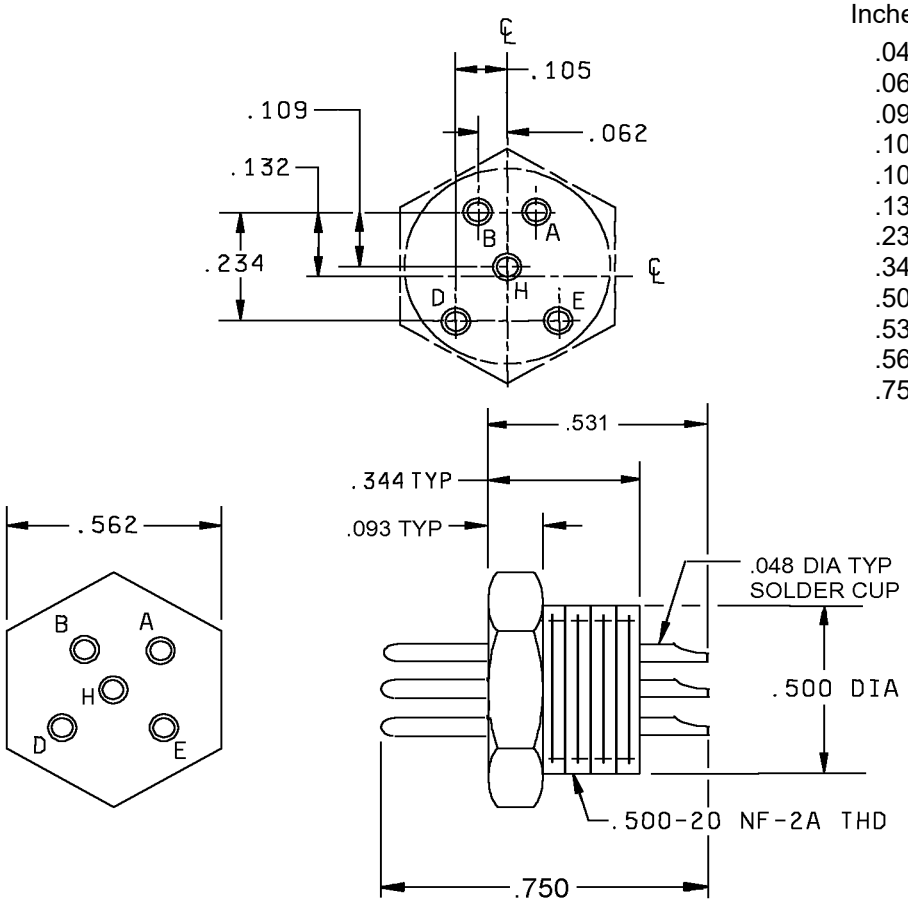


Inches	mm
.048	1.22
.085	2.16
.086	2.18
.093	2.36
.172	4.37
.344	8.74
.375	9.53
.468	11.89
.531	13.49
.750	19.05

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Unless otherwise specified, tolerances are ± .02 inch (0.51 mm) for two place decimals, and ± .015 inch (0.38 mm) for three place decimals.
4. Tolerance between any two adjacent contact centers shall be ± .004 inch (0.10 mm). Tolerance between any two contact centers, other than adjacent contacts, shall be ± .006 inch (0.15 mm).
5. Dimensions symmetrical about centerlines.
6. Contact identification may be located on either side of the contact hold but shall appear on the front and rear face of the contact insert.
7. Contact locations are shown from mating face.
8. Material: Mineral-filled melamine, type MME, in accordance with ASTM D5948.
9. Insert size 4, polarization is accomplished by pin “A” .056 inch (1.42 mm) diameter contact.
10. Without nut and lockwasher connector may be used as a plug on cable.

FIGURE 3. Insert 4 contacts.

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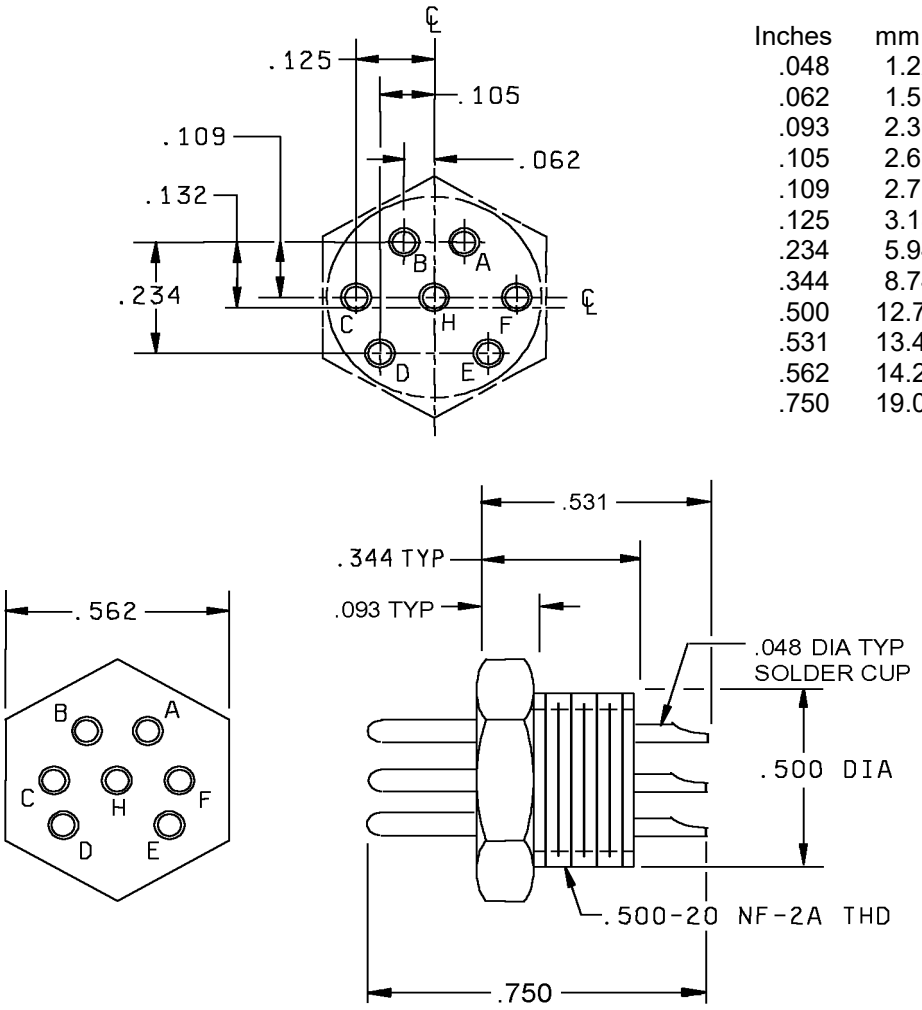


Inches	mm
.048	1.22
.062	1.57
.093	2.36
.105	2.67
.109	2.77
.132	3.35
.234	5.94
.344	8.74
.500	12.70
.531	13.49
.562	14.27
.750	19.05

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Unless otherwise specified, tolerances are ± .02 inch (0.51 mm) for two place decimals, and ± .015 inch (0.38 mm) for three place decimals.
4. Tolerance between any two adjacent contact centers shall be ± .004 inch (0.10 mm). Tolerance between any two contact centers, other than adjacent contacts, shall be ± .006 inch (0.15 mm).
5. Dimensions symmetrical about centerlines.
6. Contact identification may be located on either side of the contact hold but shall appear on the front and rear face of the contact insert.
7. Contact locations are shown from mating face.
8. Material: Mineral-filled melamine, type MME, in accordance with ASTM D5948.
9. Without nut and lockwasher connector may be used as a plug on cable.

FIGURE 4. Insert 5 contacts.

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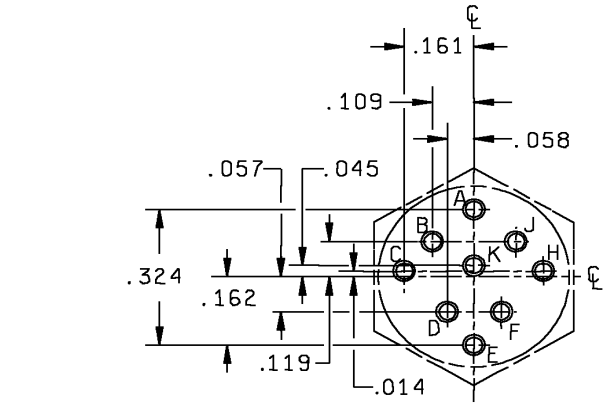
Inches	mm
.048	1.22
.062	1.57
.093	2.36
.105	2.67
.109	2.77
.125	3.18
.234	5.94
.344	8.74
.500	12.70
.531	13.49
.562	14.27
.750	19.05

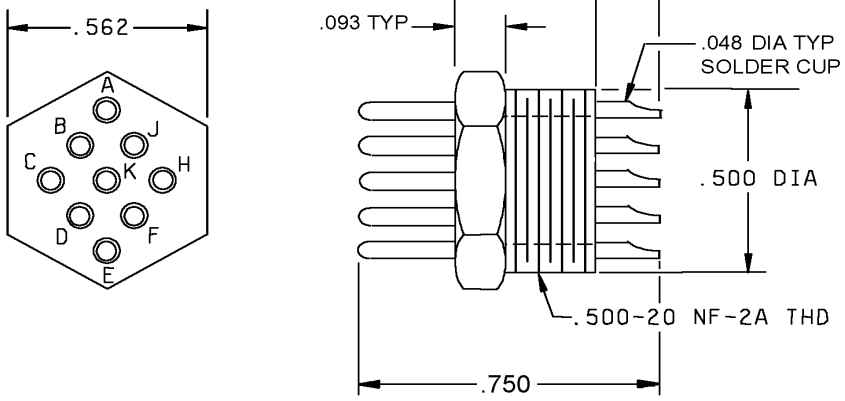
NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Unless otherwise specified, tolerances are ± .02 inch (0.51 mm) for two place decimals, and ± .015 inch (0.38 mm) for three place decimals.
4. Tolerance between any two adjacent contact centers shall be ± .004 inch (0.10 mm). Tolerance between any two contact centers, other than adjacent contacts, shall be ± .006 (0.15 mm) inch.
5. Dimensions symmetrical about centerlines.
6. Contact identification may be located on either side of the contact hold but shall appear on the front and rear face of the contact insert.
7. Contact locations are shown from mating face.
8. Material: Mineral-filled melamine, type MME, in accordance with ASTM D5948.
9. Without nut and lockwasher connector may be used as a plug on cable.

FIGURE 5. Insert 7 contacts.

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Inches	mm
.014	0.36
.045	1.14
.048	1.22
.057	1.45
.058	1.47
.093	2.36
.109	2.77
.119	3.02
.161	4.09
.162	4.11
.324	8.23
.344	8.74
.500	12.70
.531	13.49
.562	14.27
.750	19.05



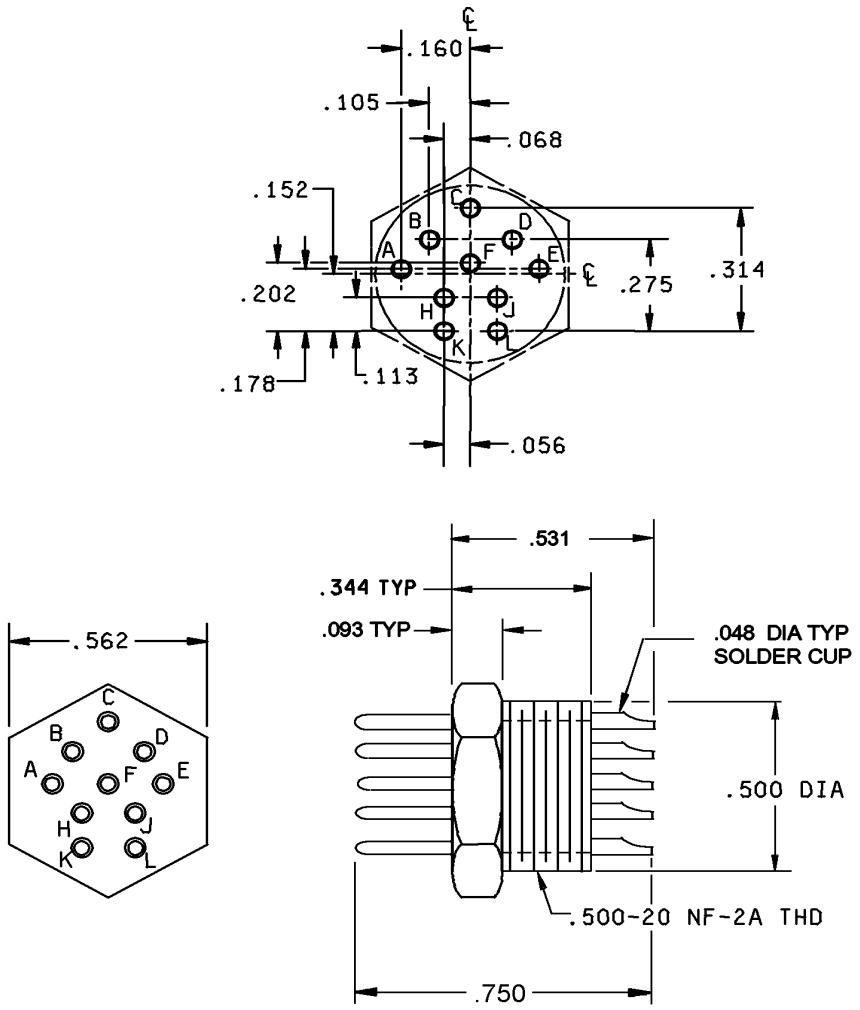


NOTES:

1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Unless otherwise specified, tolerances are ± .02 inch (0.51 mm) for two place decimals, and ± .015 inch (0.38 mm) for three place decimals.
4. Tolerance between any two adjacent contact centers shall be ± .004 inch (0.10 mm). Tolerance between any two contact centers, other than adjacent contacts, shall be ± .006 inch (0.15 mm).
5. Dimensions symmetrical about centerlines.
6. Contact identification may be located on either side of the contact hold but shall appear on the front and rear face of the contact insert.
7. Contact locations are shown from mating face.
8. Material: Mineral-filled melamine, type MME, in accordance with ASTM D5948.
9. Without nut and lockwasher connector may be used as a plug on cable.

FIGURE 6. Insert 9 contacts.

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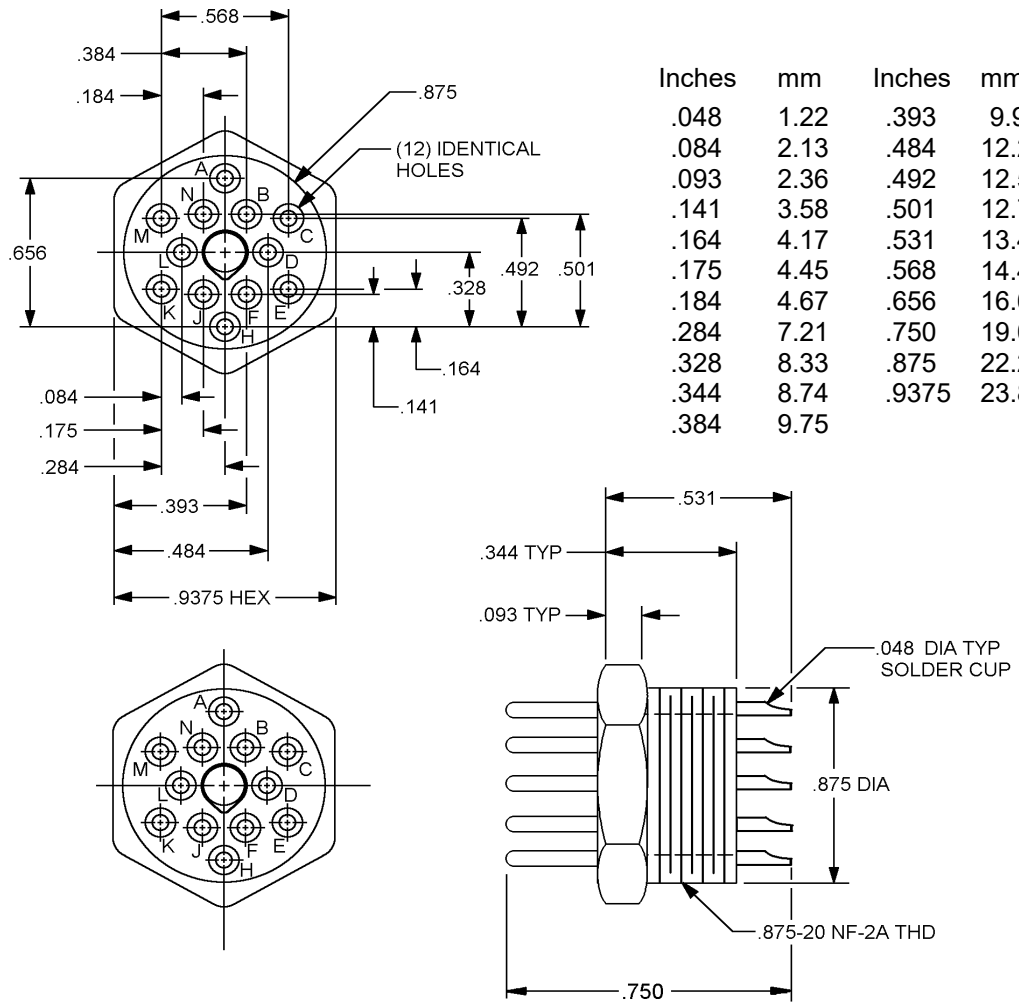


Inches	mm
.048	1.22
.056	1.42
.068	1.73
.093	2.36
.105	2.67
.113	2.87
.152	3.86
.160	4.06
.178	4.52
.202	5.13
.275	6.99
.314	7.99
.344	8.74
.500	12.70
.531	13.49
.562	14.27
.750	19.05

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Unless otherwise specified, tolerances are ± .02 inch (0.51 mm) for two place decimals, and ± .015 inch (0.38 mm) for three place decimals.
4. Tolerance between any two adjacent contact centers shall be ± .004 inch (0.10 mm). Tolerance between any two contact centers, other than adjacent contacts, shall be ± .006 inch (0.15 mm).
5. Dimensions symmetrical about centerlines.
6. Contact identification may be located on either side of the contact hold but shall appear on the front and rear face of the contact insert.
7. Contact locations are shown from mating face.
8. Material: Mineral-filled melamine, type MME, in accordance with ASTM D5948.
9. Without nut and lockwasher connector may be used as a plug on cable.

FIGURE 7. Insert 10 contacts.

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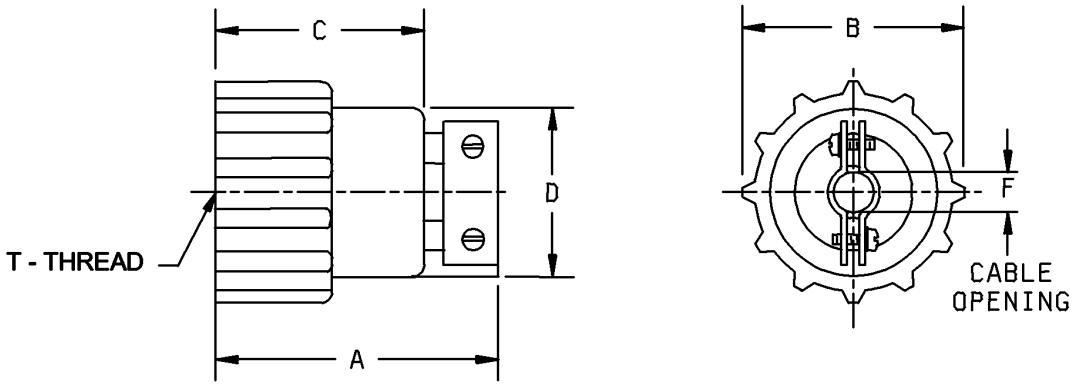


Inches	mm	Inches	mm
.048	1.22	.393	9.98
.084	2.13	.484	12.29
.093	2.36	.492	12.50
.141	3.58	.501	12.73
.164	4.17	.531	13.49
.175	4.45	.568	14.43
.184	4.67	.656	16.66
.284	7.21	.750	19.05
.328	8.33	.875	22.23
.344	8.74	.9375	23.81
.384	9.75		

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Unless otherwise specified, tolerances are ± .02 inch (0.51 mm) for two place decimals, and ± .015 inch (0.38 mm) for three place decimals.
4. Tolerance between any two adjacent contact centers shall be ± .004 inch (0.10 mm). Tolerance between any two contact centers, other than adjacent contacts, shall be ± .006 inch (0.15 mm).
5. Dimensions symmetrical about centerlines.
6. Contact identification may be located on either side of the contact hold but shall appear on the front and rear face of the contact insert.
7. Contact locations are shown from mating face.
8. Material: Mineral-filled melamine, type MME, in accordance with ASTM D5948.
9. Without nut and lockwasher connector may be used as a plug on cable.

FIGURE 8. Insert 12 contacts.

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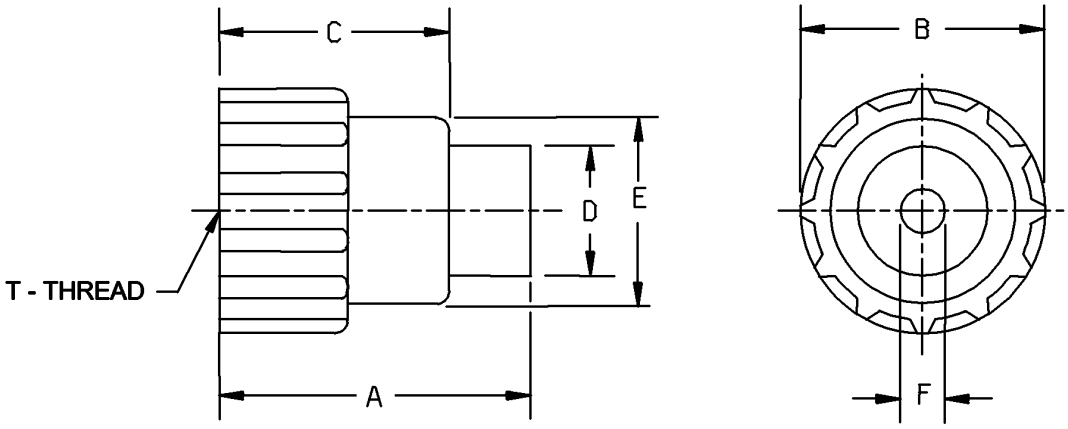
Hood designator	Connector size	A (overall length) (mm)	B (overall dia (mm)	C (mm)	D (dia) (mm)	F dia Cable clamp (mm)	T thread
C	4	.83 (21.1)	.56 (14.2)	.69 (17.5)	.44 (11.2)	.21 (5.3)	.375-24-UNF-2B
D	5, 7, 9, 10	.86 (21.8)	.66 (16.8)	.66 (16.8)	.54 (13.7)	.30 (7.6)	.500-20-UNF-2B
C	5, 7, 9, 10	.97 (24.6)	.69 ±.06 (16.8±1.5)	.78 19.8	.62±.12 (15.7±3.0)	.30 (7.6)	.500-20-UNF-2B
E	5, 7, 9, 10	.97 (24.6)	.66 (16.76)	.78 (19.8)	.55 (14.0)	.16 (4.1)	.500-20-UNF-2B

NOTES:

1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Unless otherwise specified, tolerances are ± .02 inch (0.51 mm) for two place decimals, and ± .015 inch (0.38 mm) for three place decimals.
4. Material: Glass filled diallyl phthalate in accordance with ASTM D5948, type SDG-F or mineral filled melamine-phenolic copolymer in accordance with ASTM D5948, type MME.
5. Cable clamp and screws: Brass in accordance with ASTM B36/B36M, alloy No. 6, half hard or aluminum in accordance with ASTM B209. The PIN letters C, D, and E under hood designator include the cable clamp and screws.
6. Finish: Cable clamp and screws, cadmium plate in accordance with SAE-AMS-QQ-P-416, type II, class 2, plus olive drab iridite or yellow anodize in accordance with MIL-A-8625, type II, class 2. The PIN letters C, D, and E under hood designator include the cable clamp and screws.

The PIN FIGURE 9. Hood with cable clamps.

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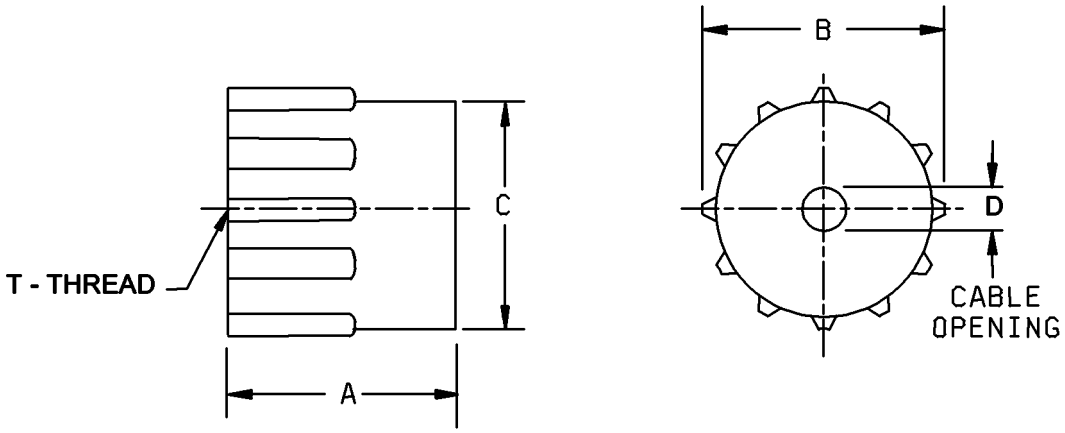
Hood designator	Connector size	A (overall length) (mm)	B (mm)	C (mm)	D (mm)	E dia (mm)	F dia Cable opening (mm)	T thread
P	4	.75 (19.1)	.56 (14.2)	.69 (17.5)	.25 (6.4)	.44 (11.2)	.16 (4.1)	.375-24-UNF-2B
P	5, 7, 9, 10	.88 (22.4)	.66 (16.8)	.66 (16.8)	.42 (10.7)	.54 (13.7)	.30 (7.6)	.500-20-UNF-2B
M	5, 7, 9, 10	.88 (22.4)	.66 (16.8)	.66 (16.8)	.27 (6.9)	.54 (13.7)	.14 (3.6)	.500-20-UNF-2B

NOTES:

1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Unless otherwise specified, tolerances are ± .02 inch (0.51 mm) for two place decimals, and ± .015 inch (0.38 mm) for three place decimals.
4. Material: Glass filled diallyl phthalate in accordance with ASTM D5948, type SDG-F or mineral filled melamine-phenolic copolymer in accordance with ASTM D5948, type MME.
5. Cable clamp and screws: Brass in accordance with ASTM B36/B36M, alloy No. 6, half hard. The PIN letters P or M under hood designator does not include the cable clamp and screws.
6. Finish: Cable clamp and screws, cadmium plate in accordance with SAE-AMS-QQ-P-416, type II, class 2, plus olive drab iridite. The PIN letters P or M under hood designator does not include the cable clamp and screws.

FIGURE 10. Plastic regular hood.

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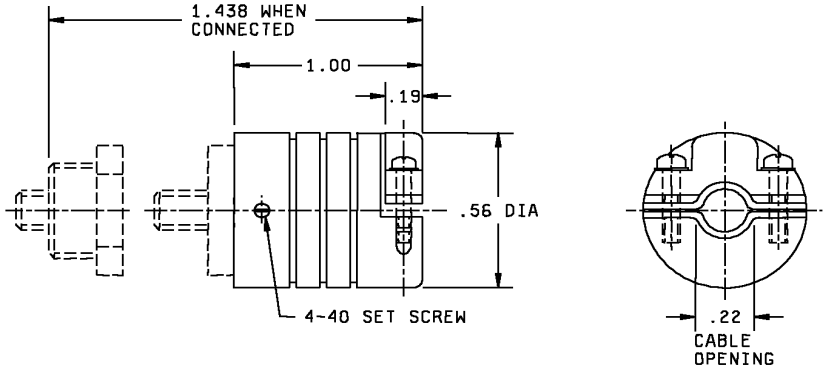


Hood designator	Connector size	A (overall length) (mm)	B (overall dia) (mm)	C (mm)	D Cable opening (mm)	T thread
S	4	.69 (17.5)	.56 (14.2)	.44 (11.2)	.16 (4.1)	.375-24-UNF-2B
S	5, 7, 9, 10	.66 (16.8)	.66 (16.8)	.54 (13.7)	.30 (7.6)	.500-20-UNF-2B
T	5, 7, 9, 10	.66 (16.8)	.66 (16.8)	.54 (13.7)	.14 (3.56)	.500-20-UNF-2B

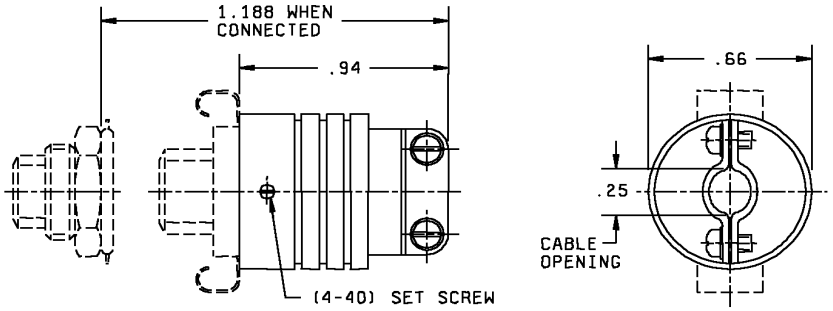
NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Unless otherwise specified, tolerances are ± .02 inch (0.51 mm) for two place decimals, and ± .015 inch (0.38 mm) for three place decimals.
4. Material: Glass filled diallyl phthalate in accordance with ASTM D5948, type SDG-F or mineral filled melamine-phenolic copolymer in accordance with ASTM D5948, type MME.
5. Cable clamp and screws: Brass in accordance with ASTM B36/B36M, alloy No. 6, half hard. The PIN letters S and T under hood designator does not include the cable clamp and screws.
6. Finish: Cable clamp and screws, cadmium plate in accordance with SAE AMS-QQ-P-416, type II, class 2, plus olive drab iridite. The PIN letters S and T under hood designator does not include the cable clamp and screws.

FIGURE 11. Plastic short hood.

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FOR SIZE 4



FOR SIZE 5,7,9,10

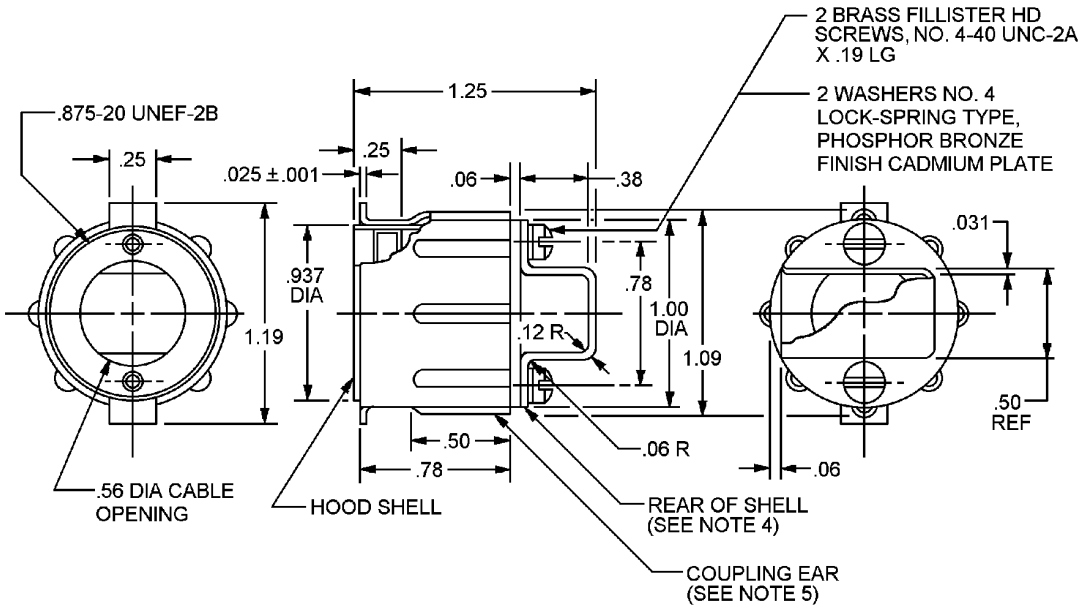
Inches	mm
.19	4.83
.22	5.59
.25	6.35
.56	14.22
.66	16.76
.94	23.88
1.00	25.40
1.188	30.18
1.438	36.53

Hood designator	Connector size	A (overall length) (mm)	B (mm)	C (mm)	D (overall dia) (mm)	E dia (mm)	F dia Cable opening (mm)	T thread
A or B	4	1.06 (26.92)	.80 (20.32)	.25 (6.35)	.56 (14.22)	.56 (14.22)	.22 (5.59)	.375-24-UNF-2B
A or B	5, 7, 9, 10	.94 (23.88)	.71 (18.03)	.25 (6.35)	.66 (16.76)	.58 (14.73)	.25 (6.35)	.500-20-UNF-2B

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Unless otherwise specified, tolerances are ± .02 inch (0.51 mm) for two place decimals, and ± .015 inch (0.38 mm) for three place decimals.
4. Tolerance between any two adjacent contact centers shall be ± .004 inch (0.10 mm). Tolerance between any two contact centers, other than adjacent contacts, shall be ± .006 inch (0.15 mm).
5. Dimensions symmetrical about centerlines.
6. Contact identification may be located on either side of the contact hold but shall appear on the front and rear face of the contact insert.
7. Material aluminum alloy in accordance with SAE-AMS-QQ-A-250/8, temper H327.
8. Finish anodize in accordance with MIL-A-8625, type I or type II.

FIGURE 12. Aluminum hood.

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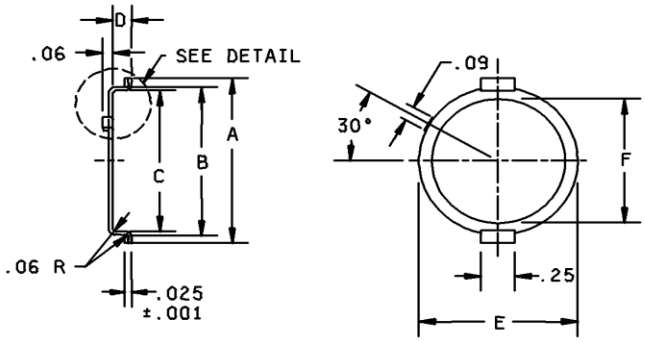


Inches	mm	Inches	mm
.001	0.03	.50	12.70
.025	0.64	.56	14.22
.031	0.79	.78	19.81
.06	1.52	.875	22.23
.12	3.05	.937	23.80
.25	6.35	1.00	25.40
.31	7.87	1.09	27.69
.38	9.65	1.19	30.23
		1.25	31.75

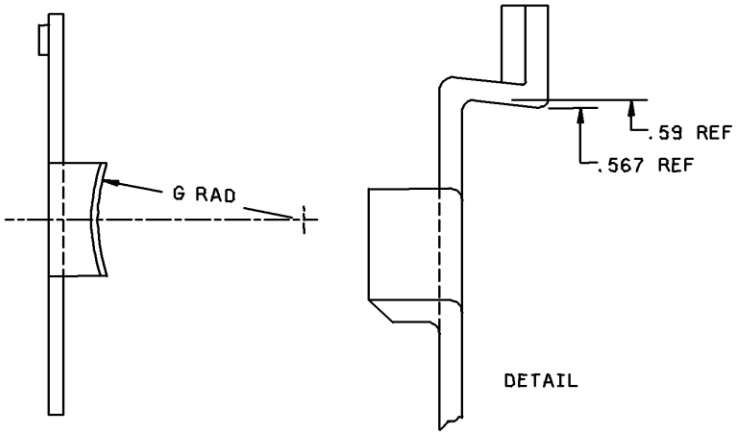
NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Unless otherwise specified tolerances are ± .02 inch (0.51 mm) for two place decimals, and ± .005 inch (0.13 mm) for three place decimals.
4. Coupling ear to be free and independent of hood shell.
5. The rear of the shell shall be made detachable for maintenance purposes.
6. Screws brass in accordance with ASTM B36/B36M, copper alloy Unified Numbering System (UNS) designator C26000, Rockwell hardness B57 to B77.
7. Material aluminum alloy in accordance with SAE-AMS-QQ-A-250/8, temper H327.
8. Finish anodize in accordance with MIL-A-8625, type I or type II.
9. Hood is to be used in conjunction with coupling ear (lock ring) and brass nut.

FIGURE 13. Right angle hood designator R (insert size 12).

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Inches	mm
.001	0.03
.025	0.64
.06	1.52
.09	2.29
.25	6.35
.567	14.40
.59	14.98

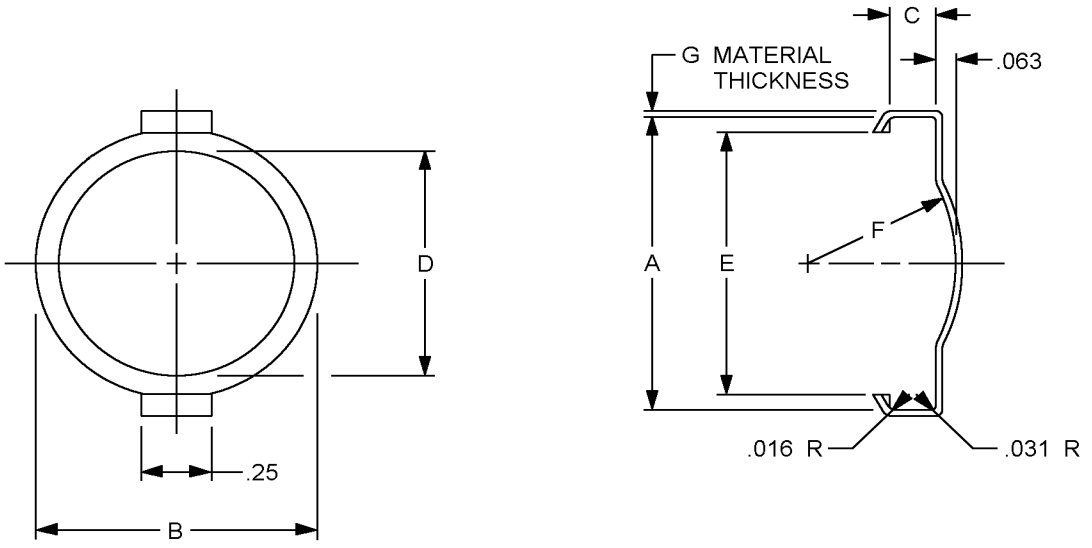


Lock type	Insert size	A (mm)	B (mm)	C ±.015 (mm)	D (mm)	E Dia (mm)	F Dia ±.010	G Rad (mm)
R	4	.62 (15.75)	.50 (12.70)	.473 (12.01)	.09 (2.29)	.47 (11.94)	.380 (9.65)	.25 (6.35)
R	5, 7, 9, 10	.72 (18.29)	.61 (15.49)	.567 (14.40)	.08 (2.03)	.66 (16.76)	.515 (13.08)	.25 (6.35)
R	12	1.12 (28.45)	.97 (24.64)	.942 (23.93)	.11 (2.79)	1.06 (26.92)	.882 (22.40)	.50 (12.70)

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Unless otherwise specified, tolerances are ± .02 inch (0.51 mm) for two place decimals, and ± .015 inch (0.38 mm) for three place decimals.
4. Materials: Brass in accordance with ASTM B36/B36M, copper alloy UNS designator C26000, Rockwell hardness B57 to B77.
5. Finish cadmium plate in accordance with SAE-AMS-QQ-P-416, type II, class 3.

FIGURE 14. Coupling-ear (lock-ring).

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Inches	mm
.010	0.25
.016	0.41
.031	0.79
.063	1.60
.250	6.35

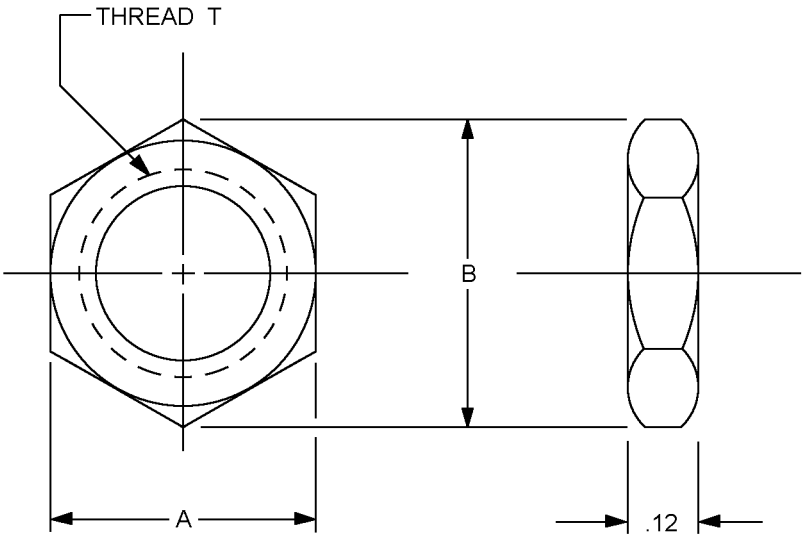
Lock type	Insert size	A (mm)	B (mm)	C (mm)	D +.010 (mm)	E (mm)	F Radius (mm)	G (mm)
S	4	.656 (16.66)	.468 (11.89)	.145 (3.68)	.380 (9.65)	.515 (13.08)	.250 (6.35)	0.016 (0.41)
S	5, 7, 9, 10	.750 (19.05)	.641 (16.28)	.145 (3.68.)	.507 (12.88)	.621 1/ (15.77)	.500 (12.70)	.010 (0.25)
S	12	1.156 (29.36)	1.046 (26.57)	.171 (4.34)	.882 (22.40)	1.031 (26.19)	.750 (19.05)	.0159 (4.04)

1/ Tolerance is ±.010 inch (0.25 mm).

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for reference purposes only.
3. Tolerance unless otherwise specified is ± .005 inch (0.13 mm).
4. Material: Phosphor bronze, in accordance with ASTM B139/B139M, copper alloy UNS designator C51000, spring temper.
5. Finish: Cadmium plate in accordance with SAE-AMS-QQ-P-416, type II, class 3.

FIGURE 15. Coupling ring, rotating (lock spring).

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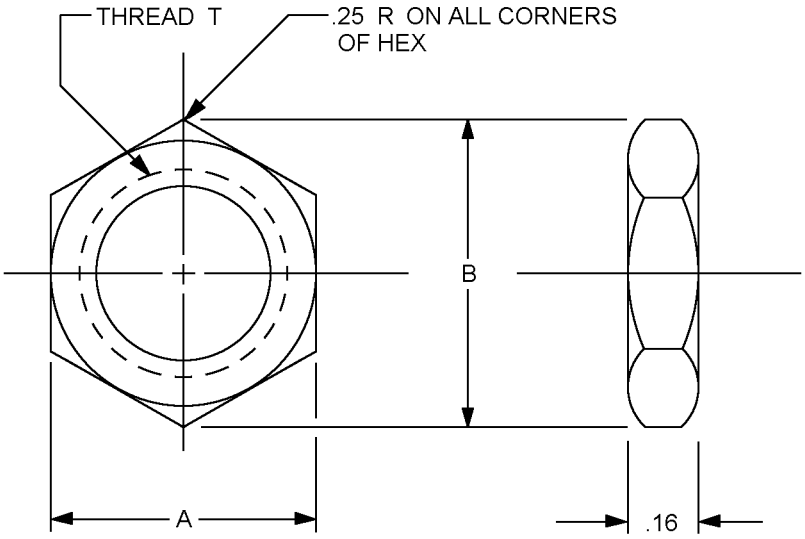
Inch	mm
.12	3.05

Nut designator	Connector size	A (mm)	B (mm)	T thread
N	4	.502 (12.75) .492 (12.50)	.56 (14.22)	.375-24 UNF-2B
N	5, 7, 9, 10	.564 (14.33) .553 (14.05)	.66 (16.8)	.500-20UNF-2B
N	12	1.12 (28.45) 1.11 (28.19)	1.22 (31.00)	.875-20 UNEF-2B

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Unless otherwise specified, tolerances are ± .02 inch (0.51 mm) for two place decimals, and ± .015 inch (0.38 mm) for three place decimals.
4. Material: Brass in accordance with ASTM B21/B21M, copper alloy UNS designator C46200 or ASTM B283/B283M, ASTM B124/B124M, copper alloy UNS designator C46400.
5. Finish: Cadmium plate in accordance with SAE-AMS-QQ-P-416, type II, class 3.

FIGURE 16. Brass nut.

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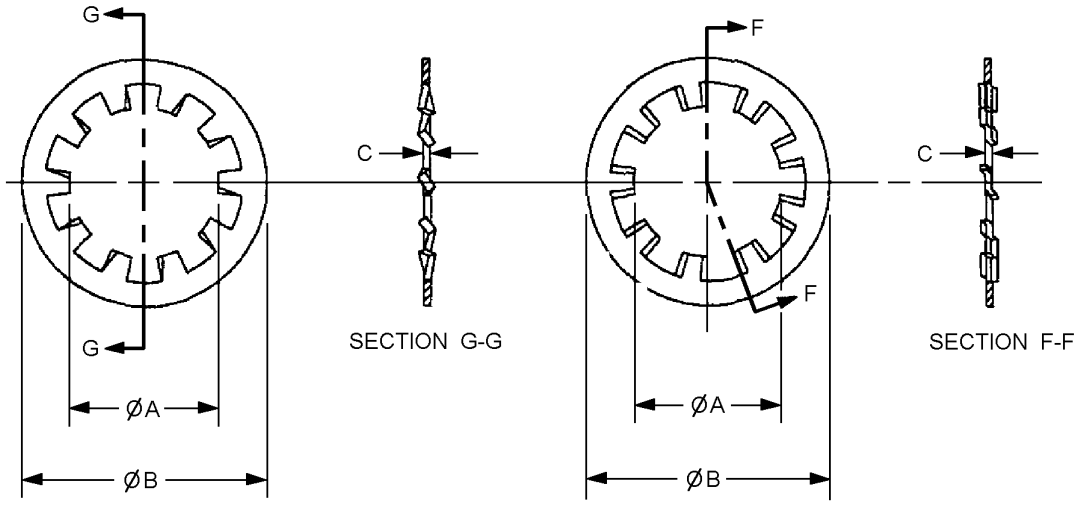
Inch	mm
.16	4.06

Nut designator	Connector size	A (mm)	B (mm)	T thread
P	4	.502 (12.75) .492 (12.50)	.56 (14.22)	.375-24 UNF-2B
P	5, 7, 9, 10	.564 (14.33) .553 (14.05)	.66 (16.8)	.500-20 UNF-2B
P	12	1.12 (28.45) 1.11 (28.19)	1.22 (31.00)	.875-20 UNEF-2B

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Unless otherwise specified, tolerances are ± .02 inch (0.51 mm) for two place decimals, and ± .015 inch (0.38 mm) for three place decimals.
4. Material: 30 percent glass fiber reinforced nylon 6\6 or equivalent.

FIGURE 17. Nylon nut.

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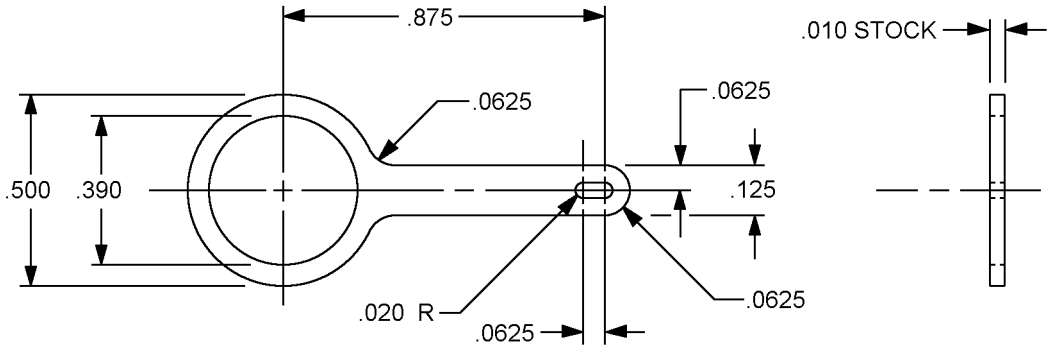
ACCEPTABLE DESIGNS

Washer PIN NASM35333- (see note 5)	Washer designator	Connector size	Size	Inside dia.		Outside dia		Thickness	
				Max	Min	Max	Min	Max	Min
42	W	4	.375	.398	.384	.692	.670	.040	.032
44	W	5, 7, 9, 10	.500	.530	.512	.900	.867	.045	.037
48	W	12	.875	.927	.894	1.410	1.364	.060	.052

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Material: Carbon steel, cadmium plated in accordance with SAE-AMS2700, type II, class 2.
4. The number of teeth shall be at the option of the manufacturer.
5. Part or Identifying Number (PIN).

FIGURE 18. Lock washer (internal tooth).

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Inches	mm
.010	0.25
.020	0.51
.0625	1.59
.125	3.18
.390	9.91
.500	12.70
.875	22.23

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Tolerance ± .016 inch (0.41 mm).
4. Material: Brass strip in accordance with ASTM B36/B36M, ½ HD, copper alloy UNS designator C28000.
5. Tumble to remove sharp edges.
6. Finish tin in accordance with SAE-AMS-P-81728, .0002 inch (0.005 mm) thick maximum.
7. Dimensions shown before plating thickness.

FIGURE 19. Ground lug.

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REQUIREMENTS

Dimensions and configuration: See figures 1 through 19 for inserts and accessories.

Insert material: Glass filled diallyl phthalate, in accordance with ASTM D5948, type SDG-F, flammability rating UL94 V-0.

Mating connector: MIL-DTL-28748/15.

Temperature range: -55°C to +125°C.

Working voltage: 300 volts rms.

Current rating: 7.5 amperes maximum.

Initial contact resistance: 0.010 ohms max.

Dielectric withstanding voltage (DWV):

Insert size 4 through 10: 1,500 VAC for one minute.

Insert size 12: 2,250 VAC for one minute.

Contact retention: 10 lbs (44.5 N) minimum.

Contacts size 20 in accordance with MIL-DTL-28748, except as specified herein.

Solder cup hole diameter:

Insert sizes 4 through 10: .048 inches (1.22 mm).

Insert size 12: .043 inches (1.09 mm).

Solder cup termination: Accepts up to size 20 AWG stranded wire.

PWB wire termination: .025 inch (0.64 mm) diameter straight solder.

Panel thickness: For 1/16-inch (1.59 mm) panel mounting receptacle use cadmium plated brass nut for insert 4 through 10, for size 12 insert use nylon nut.

Non-metallic: Non-metallic accessories consisting of hoods and hardware do not require full qualification inspection. First Article Test inspection shall consist of quality conformance inspection specified in MIL-DTL-28748, group A, examination of product

All nonmetallic material shall be selected to meet the hydrolytic reversion resistance in accordance with ASTM D570. Certification of hydrolytic stability is required. Specify material tested and water absorption report, as specified in ASTM D570. If requested test data shall be made available to the qualifying activity.

Metallic hardware: Metallic hardware consist of shells, coupling ring, coupling ear, brass nut, and ground lug. First article test inspection shall be in accordance with MIL-DTL-28748 group A.

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Lock washer shall be in accordance with Aerospace Industries Association NASM35333-42, -44, or -48, as applicable, cadmium finish (see figure 18).

The PIN shall be as shown in the following example:

M28748/16	E	5	R	A	0	0
Basic number of specification sheet	Insert size	Contact type	Lock type	Hood	Nut	Ground lug

PIN code:

Insert size	Number of contacts
A	4
B	5
C	7
D	9
E	10
F	12

Contact type
5 - Solder cup

Lock type 2/
R - lock ring
S - lock spring
0 - none

Hoods
A - Aluminum (yellow chromate)
B - Aluminum (black anodize)
C - Plastic long with cable clamp
D - Plastic short with cable clamp
E - Plastic long with cable clamp, small cable opening
M - Plastic hood long, small cable opening
P - Plastic hood long
R - Right angle (size 12)
S - Plastic short
T - Plastic short, small cable opening
0 - none

Nut
N - Brass nut
P - Nylon nut
W - Lock washer 6/
0 - none

Ground lug 3/
G -insert (size 4A only)
0 - none

NOTES:
1/ The number zero (0) is used to indicate which parts are not included.
2/ When a lock spring or a lock ring are ordered with a mounting nut, a lock washer is furnished.
3/ For insert designator A (4 contacts) only, can be use to ground any of the four contacts.
4/ To order lock washers individually use NASM35333-42, -44, or -48 as applicable, cadmium finish (see figure 18).
5/ Do to space constraints and physical size of the lock ring, lock spring, nut and ground lug, the MIL marking requirements will only be indicated on the packaging.
6/ To order accessories separately (nut, hoods, etc) use M28748/15.

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Superseded MS sheets: See table I.

TABLE I. Superseded MS sheets to M28748 designator.

Superseded MS sheet	Superseding M28748/15 designator
MS24038-1	M28748/16B50000
MS24038-2	M28748/16B50000
MS24038-3	M28748/16B50000
MS24056-1	M28748/16D50000
MS24056-2	M28748/16D50000
MS24056-3	M28748/16D50000

Permissible PINs (complete connectors): See table II.

TABLE II. Permissible PINs complete connectors.

Number of contacts	Insert designator	Contact type	Lock type	Hood	Nut	Ground lug
4	M28748/16A	5	0, R, S	0, A, B, C, P, S	0, N, P	0, G
5	M28748/16B	1, 2, 3, 4, 5	0, R, S	0, A, B, C, D, E, P, S	0, N, P	0
7	M28748/16C	1, 2, 3, 4, 5	0, R, S	0, A, B, C, D, E, P, S	0, N, P	0
9	M28748/16D	1, 2, 3, 4, 5	0, R, S	0, A, B, C, D, E, P, S	0, N, P	0
10	M28748/16E	1, 2, 3, 4, 5	0, R, S	0, A, B, C, D, E, P, S	0, N, P	0
12	M28748/16F	1, 2, 3, 4, 5	0, R, S	0, R	0, N, P	0

Qualification: Qualification is not required for this specification sheet.

First article testing (FAT): FAT shall be in accordance with MIL-DTL-28748, qualification inspection.

Amendment notations. The margins of this specification are marked with vertical lines to indicate modifications generated by this amendment. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations.

Referenced documents. In addition to MIL-DTL-28748, this document references the following:

MIL-A-8625
MIL-DTL-28748/15
ASTM B21/B21M
ASTM B36/B36M
ASTM B124/B124M
ASTM B139/B139M
ASTM B209
ASTM B283/B283M
ASTM D570
ASTM D5948

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NASM35333
SAE-AMS-QQ-A-250/8
SAE-AMS2700
SAE-AMS-QQ-P-416
SAE-AMS-P-81728

CONCLUDING MATERIAL

Custodians:
Army - CR
Navy - EC
Air Force – 85
DLA -CC

Preparing activity:
DLA - CC

(Project 5935-2018-181)

Review activities:
Army - AR, AT, AV, CR4, MI
Navy - AS, MC, OS, SH

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at https://assist.dla.mil.

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